Mind the Touch Gap: Real-to-Sim Contact Alignment for Dexterous Pen Spinning

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Abstract: Pen spinning is fast and contact-rich, making simulation transfer difficult. PenSpin [1] fine-tuned a proprioceptive student on successful hardware replays. We ask whether a small replay budget can instead align the contact observations used during simulator training. Using Isaac Gym [2] and Project SuperDex [3], we propose to learn a source-to-target contact mapper, re-distill a contact-aware student, and test it against proprio-only, raw-contact, randomized-contact, calibrated-contact, and target-fine-tuning baselines. This sim-to-sim study tests the value of contact-observation alignment without claiming to correct dynamics or establish hardware transfer.

Keywords: Dexterous manipulation, contact alignment, sim-to-sim transfer

1 Introduction

A pen spin depends on a sequence of brief, well-timed contacts: the hand has to release the pen, let it rotate, and catch it again before it falls. Simulation makes it possible to practice this motion at scale, yet even small differences in contact timing can disrupt transfer to a new environment. Dexterous in-hand learning has made considerable progress [4, 5, 6], but PenSpin [1] illustrates the remaining difficulty. It trained a privileged oracle and distilled a student from proprioception alone; the student did not transfer reliably to hardware. PenSpin improved hardware performance by replaying oracle actions, keeping successful trajectories, and fine-tuning the student. That adaptation uses little target data, but leaves the contact observations available during simulator training untouched.

We propose to use a small set of replay attempts to improve those observations instead. Previous approaches have adapted simulator parameters from target experience [7, 8, 9] or calibrated tactile simulation through replay [10]. Our proposal focuses on the contact signal received by the policy: paired rollouts teach a mapping from source link-level contact, proprioception, and past actions to the contact reported in the target. We then freeze the mapping and re-distill a student on the aligned signal. The intuition is that a student trained with target-like touch cues may make better decisions when the target reports contact differently. This does not change the source collision solver or contact dynamics. Since PenSpin's student did not use contact, a contact-aware student is an explicit addition to its pipeline; the original proprio-only student remains a baseline.

Isaac Gym [2] will provide source training and Project SuperDex [3] a target simulator with observable contact and controlled resets. In this setting, we can ask whether contact helps under friction and compliance shifts, whether limited replays suffice, and how alignment compares with target-policy fine-tuning at equal data budgets. Matched target tests and state-matched resets will help us diagnose drift between paired replays. The resulting evidence concerns observation alignment in simulation; it cannot establish hardware transfer or correct contact dynamics.

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2 Plan

We will first reproduce PenSpin's oracle and proprio-only student, add a student with binary contact on corresponding Allegro links, and check that the hand, pen, controller, and contact readings behave sensibly in SuperDex. From matched initial states, we will replay the same oracle actions in Isaac Gym and SuperDex, then train a temporal mapper using source contact, proprioception, and *preceding* actions to predict target contact. After re-distillation with the mapper frozen, the student will run in SuperDex using native target contact (Figure 1). We will compare it with the original proprio-only student, raw contact, contact randomization, simple threshold-and-delay calibration, and PenSpin-style fine-tuning. All target-data methods receive the same number of attempted replays; we will also report how many of those attempts yield successful demonstrations for fine-tuning. Tests will share target objects and initial states, with results reported across replay budgets and friction/compliance settings. Contact prediction scores and replay-state divergence will help interpret policy performance.

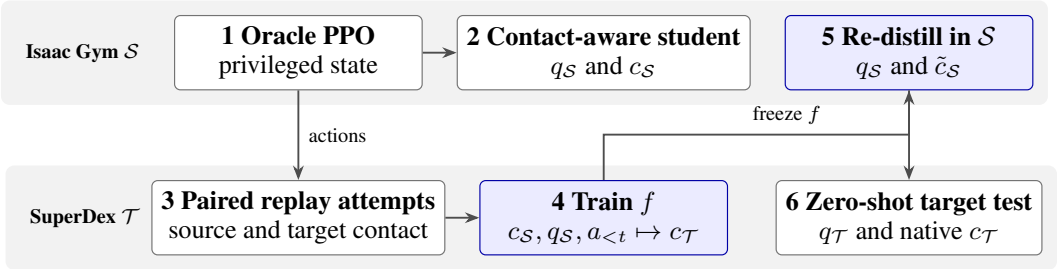


Figure 1: Proposed training and evaluation pipeline. The target simulator provides paired contact labels for the mapper and native contact for the final policy test.

- **10/13–10/30:** Reproduce PenSpin and complete a small Allegro/pen/contact replay test in SuperDex.
- **10/31–11/13:** Collect paired replays, train the contact mapper, and re-distill the student.
- **11/14–12/04:** Compare baselines at 10, 25, 50, and 100 replay attempts; measure rotations, drop time, success, contact F1, and state drift.
- **12/05–12/11:** Analyze results and prepare the final report and presentation.

If the SuperDex port proves unstable, we will first test a simpler in-hand rotation task [4] or move to a target simulator with explicit link-level contact, and report that change clearly.

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References

[1] J. Wang, Y. Yuan, H. Che, H. Qi, Y. Ma, J. Malik, and X. Wang. Lessons from learning to spin "pens". In *Conference on Robot Learning (CoRL)*, 2024. URL https://penspin.github.io/.

[2] V. Makoviychuk et al. Isaac gym: High performance GPU-based physics simulation for robot learning. In *NeurIPS Datasets and Benchmarks Track*, 2021.

[3] The Mochi Team. Project SuperDex: A unified platform for dexterous manipulation research. Open-source software, Meta (facebookresearch), 2026. URL https://github.com/facebookresearch/project_superdex.

[4] H. Qi, A. Kumar, R. Calandra, Y. Ma, and J. Malik. In-hand object rotation via rapid motor adaptation. In *Conference on Robot Learning (CoRL)*, 2022.

[5] OpenAI et al. Learning dexterous in-hand manipulation. *The International Journal of Robotics Research*, 39(1):3–20, 2020.

[6] H. Qi, B. Yi, S. Suresh, M. Lambeta, Y. Ma, R. Calandra, and J. Malik. General in-hand object rotation with vision and touch. In *Conference on Robot Learning (CoRL)*, 2023.

[7] Y. Chebotar, A. Handa, V. Makoviychuk, M. Macklin, J. Issac, N. Ratliff, and D. Fox. Closing the sim-to-real loop: Adapting simulation randomization with real world experience. In *International Conference on Robotics and Automation (ICRA)*, 2019.

[8] T. He et al. ASAP: Aligning simulation and real-world physics for learning agile humanoid whole-body skills. In *Robotics: Science and Systems (RSS)*, 2025.

[9] F. Ramos, R. Possas, and D. Fox. BayesSim: Adaptive domain randomization via probabilistic inference for robotics simulators. In *Robotics: Science and Systems (RSS)*, 2019.

[10] S. Luo, X. Huang, Z. Xu, W. Li, Z. Jiao, and C. Xiao. Blind dexterous grasping via Real2Sim2Real tactile policy learning, 2026. URL https://arxiv.org/abs/2606.11767.